Spatial Web Basics

Jan-Erik Vinje

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Co-chair of the OGC GeoPose Standards working group.
Geospatial Computing Software Engineer

ISMAR 2026

October 5 - 9, Bari, Italy

The 25th International Symposium on Mixed and Augmented Reality





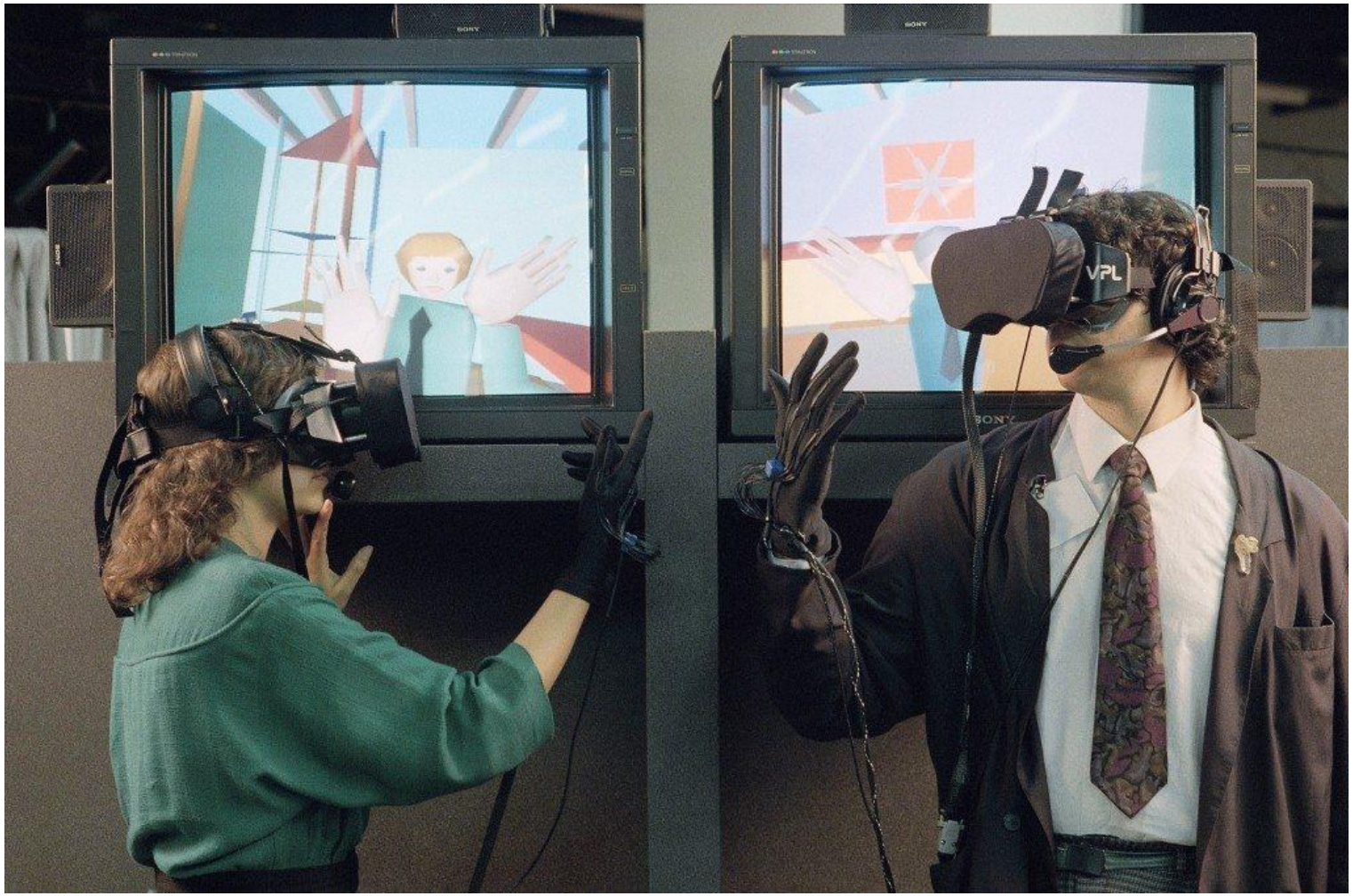
AND THE QUEST TO BUILD THE

OPEN SPATIAL WEB

1968 – Ivan Southerland



1989 – EyePhone, VPL Research



Credits Jaron Lanier (inventor of the term VR, and early VR tech-pioneer, digital rights activist)

2012 – Oculus rift




awe 2013
AUGMENTED WORLD EXPO
Produced by:
AUGMENTEDREALITY.ORG
PROXIMITY ALERT
1 2 3 4 5
> REGISTRATION OPEN
Augmented Humans in an Augmented World
> SANTA CLARA CONVENTION CENTER | JUNE 4–5, 2013

2015 – 11 yeas ago

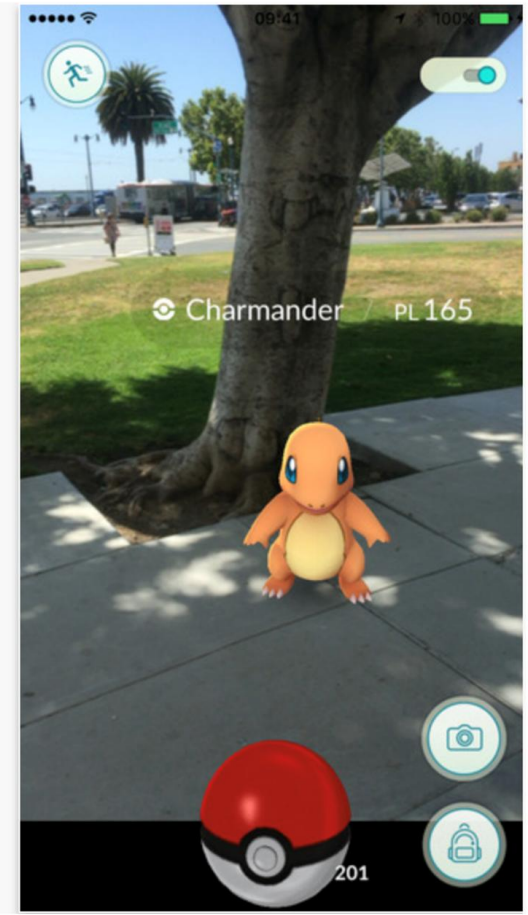


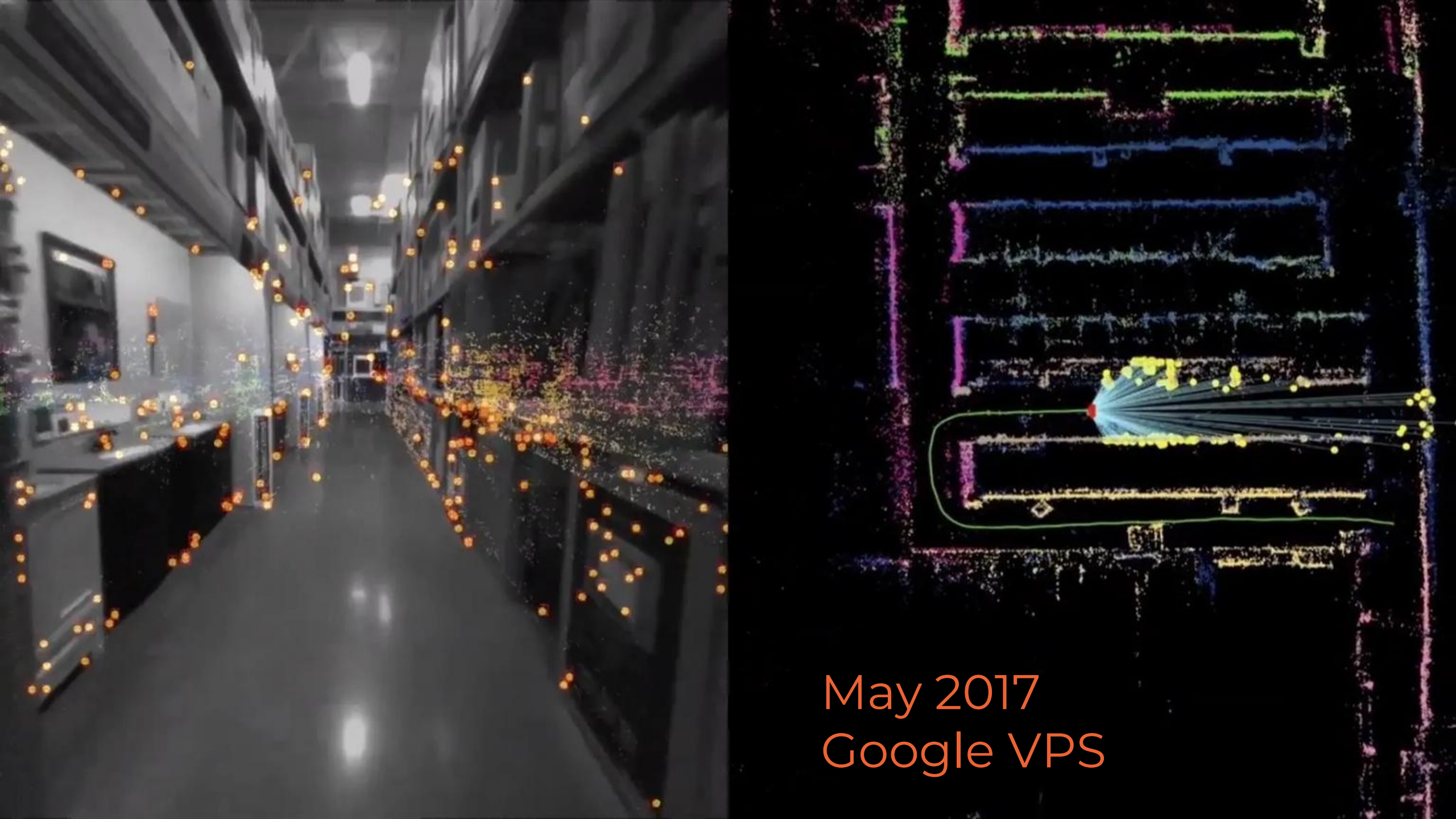


Pokémon GO

2016






May 2017
Google VPS



THE AR CLOUD

Ori Inbar
Super Ventures
September 2017



"The AR Cloud, will be the single most important software infrastructure in computing, far more valuable than facebook's Social graph or Google's page rank index."

THE AR CLOUD

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The next big thing

Exponentially improving tech

Will affect all aspects of our lives

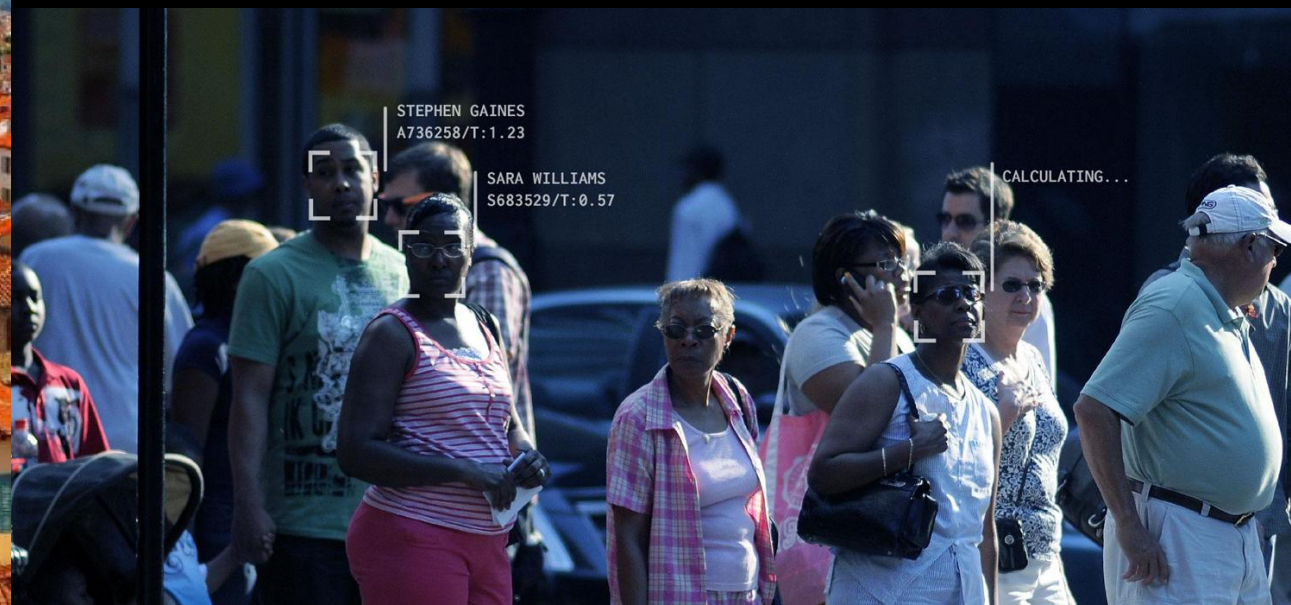
DIGITAL RENAISSANCE?

utopia



BLACK MIRROR

dystopia



A group of attendees at AWE got together around shared concerns on how this powerful emerging technology could go very wrong instead of realizing benefits for all.

Main threats:

- Lock-in to proprietary platforms
- Concentration and abuse of power
- Privacy and security risks

AWE Santa Clara June 2018



OPEN
AR CLOUD

The Open AR Cloud (OARC) is a **global non-profit organisation** founded in 2018 with the mission to drive the development of **open and interoperable spatial computing technology, data, and standards** to connect the physical and the digital worlds for the benefit of all.

AWE Munich 2018

OPEN
AR CLOUD

Mission condenced:

…to connect the **physical** and
the **digital** worlds for the
benefit of all

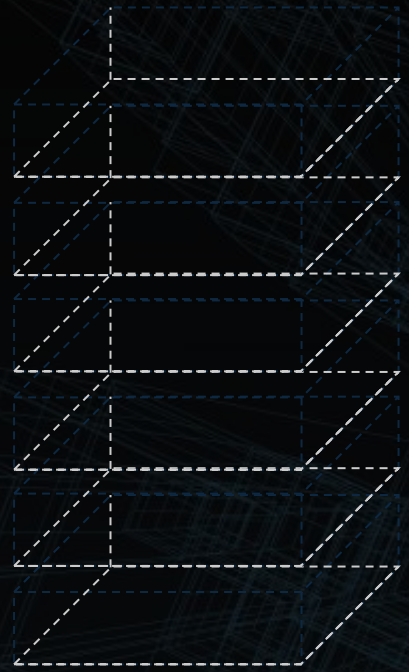
But what is **AR Cloud?**

AR Cloud: The **common cloud infrastructure** and technology ecosystem **enabling shared AR experience** where virtual spatial objects can appear and **persist in the same physical place** for all users **on every AR-platform and device.**

© 2019 Open AR Cloud Association

Layers of the **AR Cloud**

A layer can contain experiences, static content, live datastreams, services, applications/solutions etc. There needs to be a shared way of registering all such things to one or more layers.



THEMATIC LAYERS Examples

IoT

Art

Entertainment

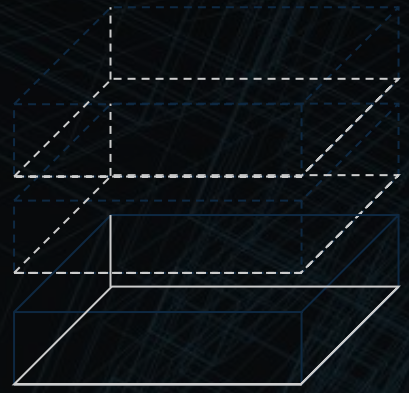
Commerce

Construction/Infrastructure

Mobility

BITS: Digital representations of the real world

ATOMS: Real world



REALITY CAPTURE LAYERS

Real-time reality layer

Transient dynamic state - restricted to only be shared locally at the edge to protect privacy, except emergency)

Static reality layer

Static persistant reality -may be shared globally

Physical reality

OPEN AR CLOUD

The reference / demonstration

Open Spatial Computing Platform (OSCP)

Standards

GeoPose

RML* / GML

Geo-glTF

Protocols

GeoPose Request & Response

RML* Subsription Query

SDS Protocol

Services

- GeoPose service
- Reality Service – provides RML*
- Spatial Discovery Service (SDS)
- Scene Understanding Service
- Shared Spatial Data Service

Client / Platform consumer

Smartphone client

AR-glasses client

RML* = Stand in for a «Reality Modeling Language» – like GML ++

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The Open Web Platform

Standards

HTML

DOM

URL / URI

JavaScript

CSS

Protocols

HTTP

TCP/IP

Services

Web-server

DNS server (nameserver)

Clients

Web browser



The collection of open (royalty-free) technologies which enables the Web. Using the Open Web Platform, everyone has the right to implement a software component of the Web without requiring any approvals or waiving license fees.

https://www.w3.org/wiki/Open_Web_Platform


SPATIAL
WEB
OPEN
AR CLOUD


Open Spatial Web
Platform

One of the biggest tech convergences we have ever seen?

AI

Smartcity

Smart mobility

Geospatial / GIS

OPEN
SPATIAL WEB
PLATFORM?

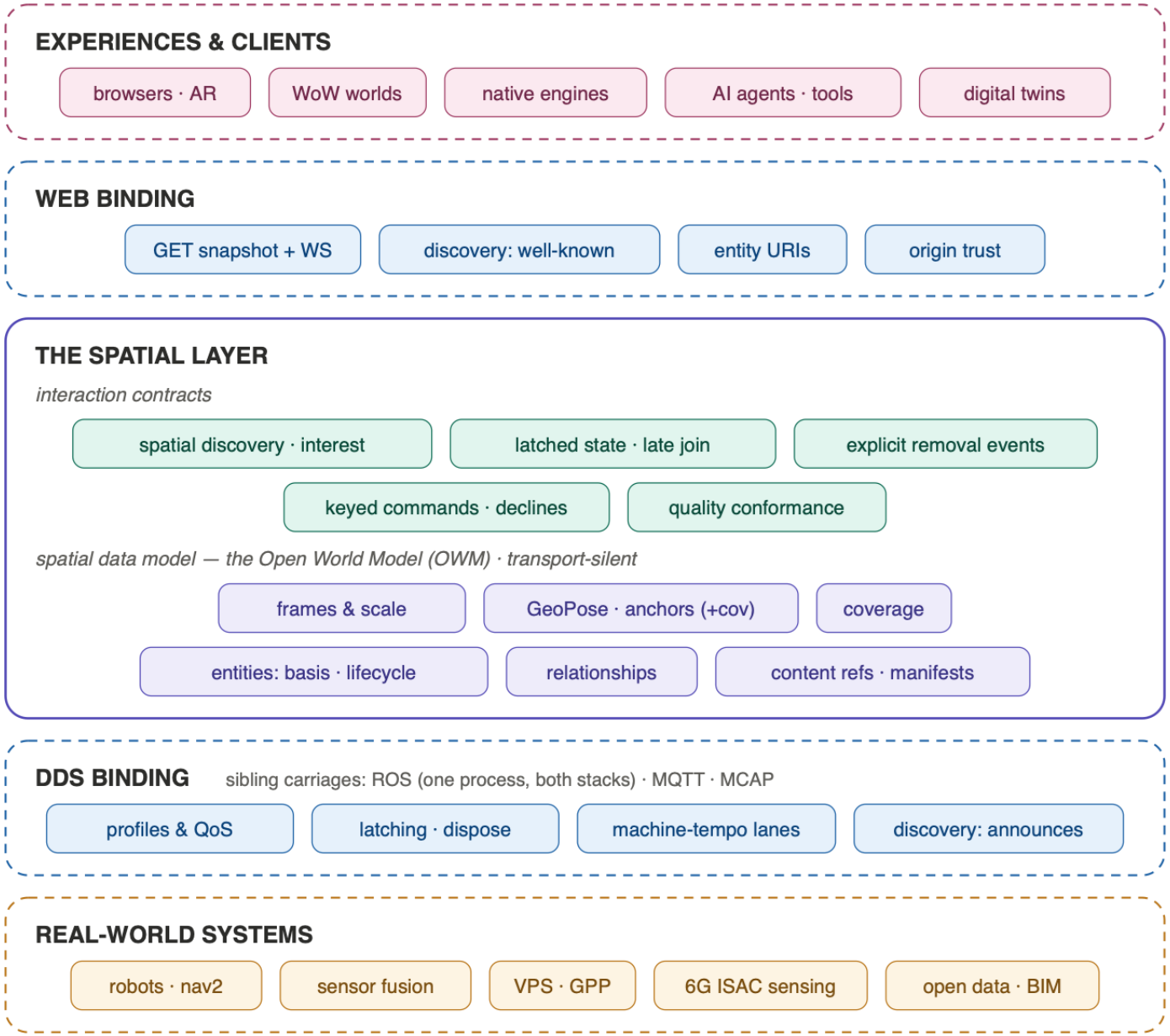
5G/Edge

SustainabilityTech

IoT

XR & AR Cloud

Stay tuned for: **Spatial DDS**



State of the art: How are things actually going in 2026?



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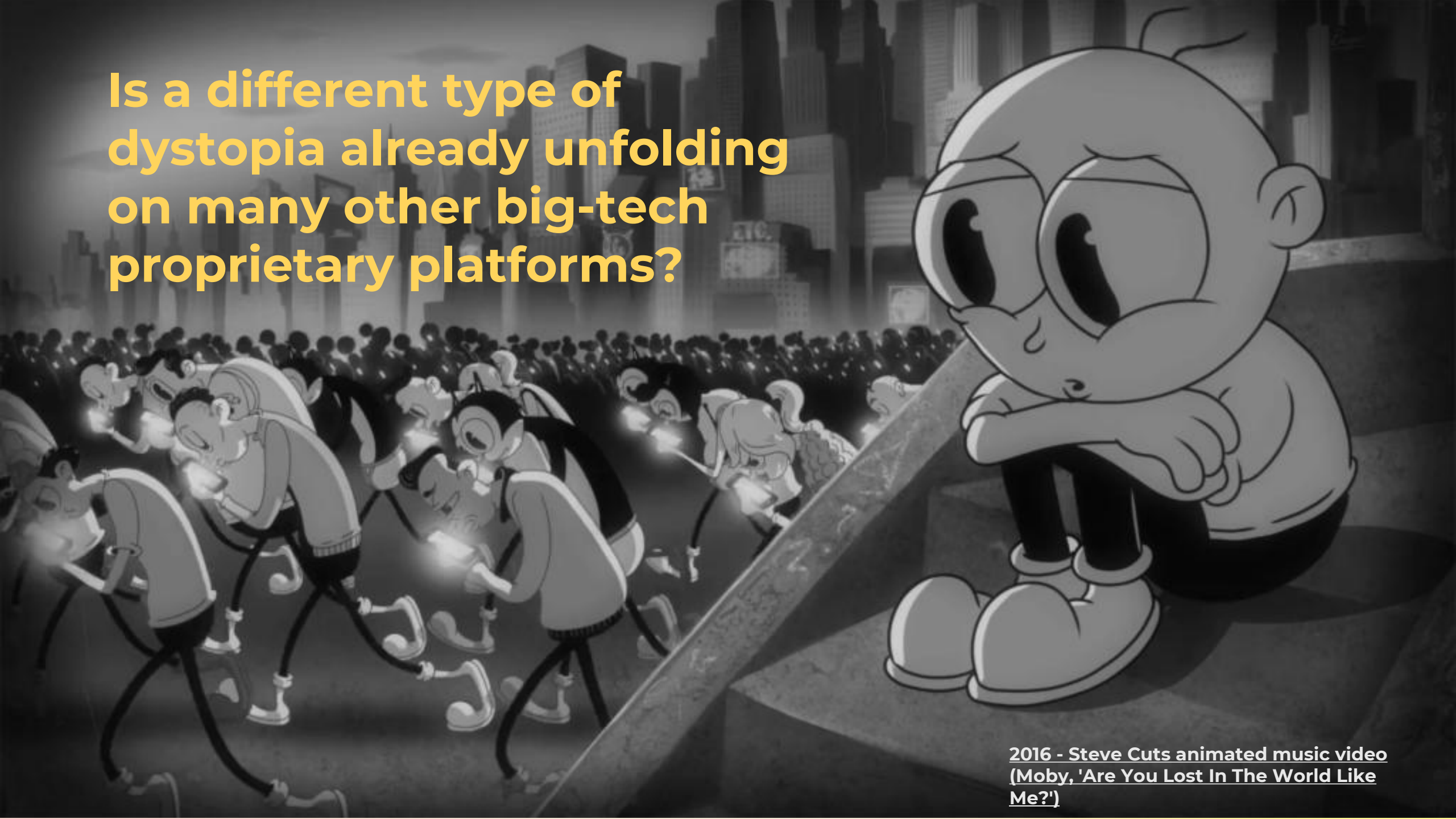


State of the art: How are things actually going in 2026?

This tech has neither brought about a Digital Renaissance nor a Black Mirror dystopia.

- No Spatial Web yet
- VPS is still very challenging
- We don´t yet have great AR Glasses
- AR Glasses are facing legal bans (privacy ++)
- Big-Tech got no success with Proprietary Platforms

Is a different type of dystopia already unfolding on many other big-tech proprietary platforms?



2016 - Steve Cuts animated music video (Moby, 'Are You Lost In The World Like Me?')

Hope and inspiration from the most unexpected place...

Utah’s Digital Choice Act

Effective July 2026

The Utah Digital Choices Access law mandates that users can take their entire social graph to new services. It requires platforms to choose a fair, non-discriminatory, interoperable protocol (like ActivityPub, AT Protocol, or DSNP) , with the state publishing qualifying technical standards.

Gradually OARC volunteers has been building something hopeful

Some highlights

2019: Started Implementing first version of the **Open Spatial Computing Platform**

2020: Started OGC Standards Working Group on **GeoPose** and initiated **city scale OSCP testbeds**

2021: Got NGI-Atlantic funding – build first **Open Source OSCP clients** and did our first exploration towards Open Source VPS

2022: Co-founded the Metaverse Standards Forum and became key contributor.

2025: Released first working Open Source VPS – on Github. Started development of RML and Spatial DDS

2026: Spatial DDS still early but has matured and is seeking SDO “home”. Increased interest from both academia and the spatial computing industry.

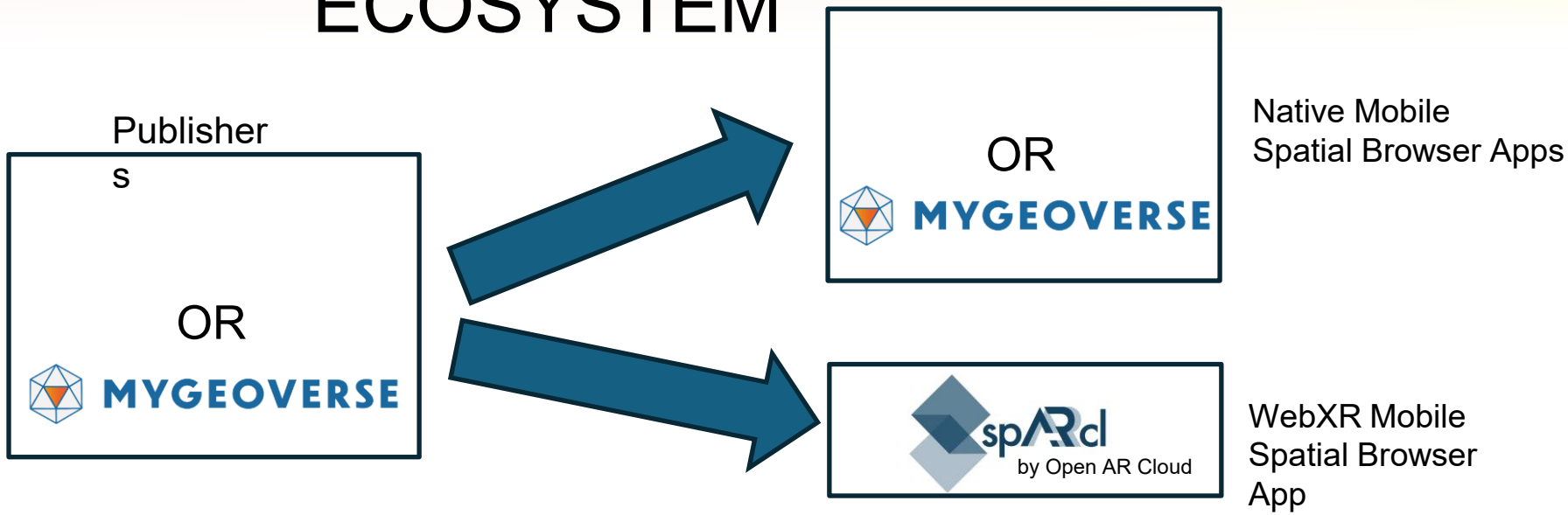
Choose to be on the right side of history!

Join the resistance against big-tech platform monopolies!

Build the open standards and protocol based spatial computing infrastrurcture for the future we actually want.

openarcloud.org

TODAY's DEMO ECOSYSTEM



VPS Service Providers

Open VPS by Open AR Cloud

IMMERSAL part of Hexagon

MultiSet

SpatialDDS by Open AR Cloud

OSCP by Open AR Cloud

GeoPose

OGC PoI